LESHEN ZHANG

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EDUCATION

Fudan University 09/2023 – 07/2027 (expected)
Bachelor's Degree in Chemistry | GPA: 3.88/4.00 *Shanghai, China*
Advisor: Prof. Gengfeng Zheng

PUBLICATIONS

‡EQUAL CONTRIBUTION, *CORRESPONDING AUTHOR

[J.1] **Leshen Zhang**, Guangsheng Liu, Xiao Ma, Abdulrahman Allangawi, Wan-Lu Li*. **Defect-Mediated Regulation of Interfacial Hydrophobic Transport via Cavitation Thermodynamics**.
Chem. Sci. **2026**, 17, 15735–15744. [PDF] [DOI] (Sole First Author)

[J.2] Guangsheng Liu‡, Xiao Ma‡, **Leshen Zhang**‡, Jenedith Pascasio, Jonathan Yang, Yuxiang Chen, Wan-Lu Li*. **CatGo: Bridging CLI Coding Agents with Interactive Structure and Workflow Management for Computational Chemistry**.
Digit. Discov. **2026**, 5, 3825–3849. [PDF] [DOI] (Co-First Author, Rank Third)

[J.3] Shuya Hao‡, Maoyin Wang‡, **Leshen Zhang**, Ximeng Lv, Chen Peng, Yuhang Huang, Pei Yuan*, Qing Han*, Gengfeng Zheng*. **Switching Photocatalytic Methane Oxidation Toward Ethanol by Tuning Spin States**.
Angew. Chem. Int. Ed. **2025**, 64, e202510241. [PDF] [DOI] (Third Author)

PREPRINTS & PAPERS UNDER REVIEW

‡EQUAL CONTRIBUTION, *CORRESPONDING AUTHOR

[S.1] **Leshen Zhang**‡, Xujian Wang‡, Taoyu Niu, Xuchao Zhou, Xiao Ma, Guangsheng Liu, Olexandr Isayev*, Junmei Wang*, Wan-Lu Li*. **Conformation-Responsive ML/MM Charges in PBSA for Protein–Ligand Affinity Ranking**.
To be submitted. (Co-First Author, Rank First)

[S.2] Guangsheng Liu‡, **Leshen Zhang**‡, Xiao Ma, Abdulrahman Allangawi, Huabin Zhang*, Wan-Lu Li*. **Local-Environment Populations Govern Size-Dependent OER Activity in Supported Subnanometer IrO_x Clusters**.
Adv. Funct. Mater., under review. [PDF] [DOI] (Co-First Author, Rank Second)

[S.3] Xiao Ma, **Leshen Zhang**, Guangsheng Liu, Abdulrahman Allangawi, Huabin Zhang*, Wan-Lu Li*. **Surface Geometry Governs Cation Effects via Nonlocal Solvent Gating**.
Angew. Chem. Int. Ed., under review. [PDF] [DOI] (Second Author)

[S.4] Guangsheng Liu‡, Xiao Ma‡, **Leshen Zhang**‡, Xujian Wang, Abdulrahman Allangawi, Shaowei Li, Wan-Lu Li*. **Intercalation-Modulated Electronic Regulation of Nitrogen Reduction on Doped MoS_2**.
To be submitted. [PDF] [DOI] (Co-First Author, Rank Third)

[S.5] Guangsheng Liu‡, **Leshen Zhang**‡, Xiao Ma, I-Shan Tsai, Jinyi Zhang, Wan-Lu Li*. **ChemMate: An Autonomous Research Mate for the Full Computational Chemistry Research Lifecycle, from Ideation to Manuscript**.
To be submitted. [PDF] [DOI] (Co-First Author, Rank Second)

[S.6] Tingsong Li‡, Jifeng Wu‡, Abdulrahman Allangawi‡, **Leshen Zhang**, Wan-Lu Li*, Xiangyun Xiao, Miao Hu, Mohd Adnan Khan, Rashed Aleisa, Shouwei Zuo, Heng Guo, Kuo-Wei Huang*, Ying Zhou*, Huabin Zhang*, Magnus Rueping*. **Interfacial Proton Relay Enables CO_2-to-Methanol Electroreduction on Layered Cobalt Phthalocyanine**.
J. Am. Chem. Soc., under review. (Fourth Author)

[S.7] Xuchao Zhou, Xujian Wang, Jiahao Xie, **Leshen Zhang**, Junmei Wang*. **Leveraging Machine Learning Interatomic Potentials into Fitting Fast and Accurate AMBER-Compatible Precise Force Field**.
Chem. Sci., under review. [PDF] [DOI] (Fourth Author)

[S.8] Yunzhi Wang‡, Chengyang Feng‡, Lijie Wang‡, Yang Cheng, **Leshen Zhang**, Yinglu Jia, Lingyun Zhao, Abdulrahman Allangawi, Weiwei Li*, Wan-Lu Li*, Martin Heeney, Magnus Rueping, Omar F. Mohammed*, Huabin Zhang*. **Overcoming Excitonic Bottlenecks via Molecular Intercalation into Donor Domains for NIR-Driven Hydrogen Evolution**.
Nat. Commun., under review. (Fifth Author)

SELECTED RESEARCH EXPERIENCE

I. First-Principles Calculations for Materials and Catalysis

(a) Surface Reconstruction and Interfacial Structure

- **Defect-Mediated Regulation of Interfacial Hydrophobic Transport via Cavitation Thermodynamics**
  [J.1] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Revealed that surface defects regulate interfacial hydrophobic transport by tuning the local cavitation free energy, establishing cavitation thermodynamics as a descriptor linking interfacial water structure to reactivity

- **Local-Environment Populations Govern Size-Dependent OER Activity in Supported Subnanometer IrO_x Clusters**
  [S.2] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Formulated a constant-potential ensemble-sampling methodology that maps site-specific overpotentials and activation barriers across cluster sizes and configurations, bridging local site motifs with microkinetics to identify the ideal size window for Ir clusters
- **Surface Geometry Governs Cation Effects via Nonlocal Solvent Gating**
  [S.3] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Showed that Au surface geometry governs alkali-cation effects in CO_2 reduction: interfacial water reverses the cation effects and propagates them nonlocally to pristine terraces, explaining the volcano-shaped cation dependence on polycrystalline Au
- **Site-Resolved C–C Coupling on Restructured Cu Electrodes (ongoing)**
  *PI: **Prof. Anastassia N. Alexandrova**, UC Los Angeles, USA*
  Investigating how kinetically trapped, adsorbate-roughened Cu electrodes catalyze C–C coupling during CO_2 reduction, and using site-resolved KMC simulation to identify the classes of restructured Cu sites that dominate the coupling rate

(b) Catalytic Mechanisms and Multiscale Modeling

- **Switching Photocatalytic Methane Oxidation Toward Ethanol by Tuning Spin States**
  [J.3] *PI: **Prof. Gengfeng Zheng**, Fudan University, China*
  Experiment: developed a Zn–O–Fe catalyst that selectively photo-oxidizes CH_4 to ethanol
  Computation: revealed that the medium Fe spin state preferentially stabilizes $\bullet CH_2OH$ over $\bullet OCH_3$, explaining why the catalyst with medium-spin polarization favors ethanol
- **Intercalation-Modulated Electronic Regulation of Nitrogen Reduction on Doped MoS_2**
  [S.4] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Developed a screening strategy that excludes unstable catalysts before DFT and distilled an interpretable SISSO descriptor governing NRR activity and selectivity against HER
- **Multiscale Modeling of PEM Water-Electrolysis OER: From DFT to Device**
  *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Bridged DFT, microkinetic modeling, and mass-transfer simulation in a multiscale model, showing that catalyst morphology governs local mass transport and thereby the apparent activity

II. AI-Assisted Scientific Software

- **CatGo: Bridging CLI Coding Agents with Interactive Structure and Workflow Management for Computational Chemistry**
  [J.2] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Established an open-source platform unifying the structure → DFT → HPC → analysis chain, and bridged AI coding agents to its workbench, enabling high-throughput calculations
- **ChemMate: A Conversational Research Collaborator for Computational Chemistry**
  [S.5] *PI: **Prof. Wan-Lu Li**, UC San Diego, USA*
  Established an autonomous research collaborator that runs entire projects from ideation to manuscript under human gates, demonstrated in five campaigns with reproducible results

III. Molecular Modeling: Electrostatics in Enzyme Design

- **Conformation-Responsive ML/MM Charges in PBSA for Protein–Ligand Affinity Ranking**
  [S.1] *PI: **Prof. Wan-Lu Li**, UC San Diego, and **Prof. Junmei Wang**, University of Pittsburgh, USA*
  Developed the PCME method for ML/MM embedding, which feeds conformation-resolved ML charges of the ligand and binding site into the Poisson–Boltzmann solver, accurately predicting binding affinities with high computational efficiency

- **Leveraging Machine Learning Interatomic Potentials into Fitting Fast and Accurate AMBER-Compatible Precise Force Field**
  [S.7] *PI:* ***Prof. Junmei Wang****, University of Pittsburgh, USA*
  Established an MLIP-driven workflow that builds AMBER-compatible force fields, matching QM-refined accuracy in hydration free energies at far lower cost and extending to noncanonical residues and metal sites in enzymes

SKILLS

Experimental Skills

- Inorganic/organic synthesis; characterization by GC, HPLC, UV-Vis, FT-IR, and NMR
- Electro- and photocatalytic testing with H-cells, flow cells, and membrane electrode assemblies

Theoretical Chemistry Skills

- **First principles:** DFT, GC-DFT, AIMD, and enhanced sampling (VASP, CP2K, Gaussian)
- **Molecular simulation:** ML/MM, ML-MD (Amber)

Machine Learning Skills

- **Models:** graph, convolutional, and recurrent neural networks; ML interatomic potentials
- **Applications:** MLIP training, interpretable descriptor discovery, and molecular classification

AI-Assisted Software Development

- **Research software:** co-developed CatGo (GitHub: Hello-QM/catgo-LRG), ChemMate, and agent-workspace plugins, architected and built with AI coding agents; independently developed SPARK (GitHub: WanluLigroupUCSD/SPARK), an on-the-fly kinetic Monte Carlo (KMC) package
- **Agent design:** human-in-the-loop agent tools (preview → approval → apply)

HONORS AND AWARDS

- NSFC Grant — Pilot Program for Young Students Basic Research (Undergraduate), National Natural Science Foundation of China 09/2026 – 07/2027
- Fudan University 'Syensqo' Scholarship 11/2025
- Fudan University 'Ruiqing' Scholarship 11/2024
- Fudan University & Chemours 'Future of Chemistry' Scholarship 11/2024
- Second Prize, 36th China Chemistry Olympiad (National Final) 11/2022

SCIENTIFIC PRESENTATIONS

Oral Presentations

- "Defect-Mediated Regulation of Hydrophobic Transport via Cavitation Thermodynamics" 12/2026 (expected)
  Chinese Chemical Society, The 16th National Conference of Theoretical and Computational Chemistry, Shenzhen, China
- "Spin Polarization in Photocatalysis" 11/2025
  Fudan University, Undergraduate Student Forum 2025 on Chemistry, Shanghai, China

Poster Presentations

- "Switching Photocatalytic Methane Oxidation Toward Ethanol by Tuning Spin States" 11/2025
  University of Chinese Academy of Sciences, Graduate Student Forum 2025 on Energy Catalysis, Dalian, China

PROFESSIONAL SERVICE

Peer Reviewer, *J. Colloid Interface Sci.*, 5 completed reviews 2024 – 2026